HR Attrition Analysis - Project Report

Project Overview

Dataset: IBM HR Analytics (1,470 employees)
Tools Used: Excel, MySQL, Power BI
Objective: Find key drivers of employee attrition and recommend solutions

About the Data

- 1,470 employee records
- 35 original columns
- 3 new columns created:
  AttritionFlag, AgeGroup, TenureBucket
- Overall attrition rate: 16.12%

Key Finding 1 — Department

Sales department has the highest attrition rate at 21% — nearly 5% above the company average of 16.12%

Impact: 1 in 5 Sales employees leaves every year

Key Finding 2 — Overtime

Employees working overtime are 3x more likely to leave

Overtime Yes: 30.53% attrition
Overtime No: 10.44% attrition

Impact: Overtime is the single strongest driver of attrition in this dataset

Key Finding 3 — Tenure

New employees in their first 2 years
churn at 31% — highest of all groups

0-2 yrs: 31% ← danger zone

3-5 yrs: 17%
6-10 yrs: 11%
10+ yrs: 8%

Impact: Company loses most employees
before they become fully productive

Key Finding 4 — Salary

Employees earning below ₹2,800/month
are at highest financial risk of leaving

Avg income of employees who LEFT: ₹4,787
Avg income of employees who STAYED: ₹6,833

Impact: ₹2,000 salary gap between
leavers and stayers

Recommendations

1. Review overtime policy in Sales dept
   — hire additional staff to reduce
   mandatory overtime

2. Strengthen onboarding program for
   new joiners in first 2 years
   — assign mentors, set 90-day goals

3. Salary review for employees earning
   below ₹2,800/month
   — benchmark against industry standards

4. Launch targeted retention program
   for Sales department specifically
— quarterly check-ins, career growth plan

Tools & Techniques Used

Excel:
- Converted raw CSV to Excel Table
- Created 3 helper columns using IF formulas
- Removed duplicates, cleaned text columns
- Built 3 Pivot Tables with charts
- Applied conditional formatting for outliers

SQL (MySQL):
- Created database and loaded 1,470 rows
- Wrote 4 analysis queries
- Used GROUP BY, CASE WHEN, window functions
- Used RANK() to identify highest risk segments

Power BI:
- Connected to CSV data source
- Built star schema with 4 dimension tables
- Created 7 DAX measures
- Built 6 interactive visuals
- Used AI Key Influencers visual
- Built Decomposition Tree for drill-down